

An Analysis of Graph attention network

Graph attention network

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

Graph attention network -- Part 1

where $\mathbf{|||}$ denotes vector concatenation, $\mathbf{0}$ is a vector of zeros, Θ is a matrix of learnable parameters, GRU is a GRU cell, and l denotes the sequence index. In a GGS-NN, the node representations are regarded as the hidden states of a GRU cell. The initial node features $\mathbf{x}_u^{(0)}$ are zero-padded up to the hidden state dimension of the GRU cell. The same GRU cell is used for updating representations for each node.

Graph attention network -- Part 2

Water distribution systems can be modelled as graphs, being then a straightforward application of GNN. This kind of algorithm has been applied to water demand forecasting, interconnecting District Metered Areas(DMAs) to improve the forecasting capacity. Other application of this algorithm on water distribution modelling is the development of metamodels.

Graph attention network -- Part 3

Message passing layers are permutation-equivariant layers mapping a graph into an updated representation of the same graph. Formally, they can be expressed as message passing neural networks (MPNNs). Let $G = (V, E)$ be a graph, where V is the node set and E is the edge set. Let N_u be the neighbourhood of some node $u \in V$. Additionally, let $\mathbf{x}_u$ be the features of node $u \in V$, and $\mathbf{e}_{uv}$ be the features of edge $(u, v) \in E$. An MPNN layer can be expressed as follows:

Graph attention network -- Part 4

$$\mathbf{H} = \sigma \left(\mathbf{D}^{-1/2} \mathbf{A} \mathbf{D}^{-1/2} \mathbf{X} \Theta \right)$$
$$= \sigma \left(\tilde{\mathbf{D}}^{-\frac{1}{2}} \tilde{\mathbf{A}} \tilde{\mathbf{D}}^{-\frac{1}{2}} \tilde{\mathbf{X}} \mathbf{\Theta} \right)$$

Graph attention network -- Part 5

where $i = \text{top}_k(\mathbf{y})$ is the subset of nodes with the top-k highest projection scores, $\odot$ denotes element-wise matrix multiplication, and $\text{sigmoid}(\cdot)$ is the sigmoid function. In other words, the nodes with the top-k highest projection scores are retained in the new adjacency matrix $\mathbf{A}'$. The $\text{sigmoid}(\cdot)$ operation makes the projection vector $\mathbf{p}$ trainable by backpropagation, which otherwise would produce discrete outputs.

Graph attention network -- Part 6

Social networks are a major application domain for GNNs due to their natural representation as social graphs. GNNs are used to develop recommender systems based on both social relations and item relations.

Graph attention network -- Part 7

To represent an image as a graph structure, the image is first divided into multiple patches, each of which is treated as a node in the graph. Edges are then formed by connecting each node to its nearest neighbors based on spatial or feature similarity. This graph-based representation enables the application of graph learning models to visual tasks. The relational structure helps to enhance feature extraction and improve performance on image understanding.

Graph attention network -- Part 8

Scalability and distributed training Training GNNs on large graphs presents computational challenges that differ substantially from those in other neural network architectures. Two broad problem settings arise: training on a single very large graph (such as a social network with billions of nodes), and training on collections of large individual graphs (such as molecular systems requiring higher-order interaction modeling). For the single large graph setting, sampling-based methods reduce memory requirements by operating on subgraphs rather than the full graph. Cluster-GCN partitions the graph into subgraphs using graph partitioning algorithms, training the model on each partition in sequence. GraphSAINT instead samples subgraphs stochastically during training, constructing unbiased estimators for the full graph loss. Distributed frameworks such as DistDGL extend training to multi-machine settings, managing the inter-machine communication required for neighborhood aggregation when a graph's nodes are distributed across machines. For tasks such as atomic simulations, a different bottleneck arises: GNN architectures that model higher-order

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interactions between triplets or quadruplets of atoms are memory-intensive at the level of individual graphs, making standard data parallelism — where each GPU processes a separate sample — insufficient when individual graphs exceed single-device memory. Graph Parallelism addresses this by distributing a single input graph across multiple GPUs, partitioning nodes and edges across devices rather than distributing samples across devices.. This approach enabled GNNs with hundreds of millions to billions of parameters to be trained on atomic systems that would otherwise exceed single-device memory, and has been applied to the development of large-scale universal interatomic potentials.

Graph attention network -- Part 9

GNNs are used as fundamental building blocks for several combinatorial optimization algorithms. Examples include computing shortest paths or Eulerian circuits for a given graph, deriving chip placements superior or competitive to handcrafted human solutions, and improving expert-designed branching rules in branch and bound.

Graph attention network -- Part 10

Attention in Machine Learning is a technique that mimics cognitive attention. In the context of learning on graphs, the attention coefficient α_{uv} measures how important node v is to node u . Normalized attention coefficients are computed as follows:

Graph attention network -- Part 11

When viewed as a graph, a network of computers can be analyzed with GNNs for anomaly detection. Anomalies within provenance graphs often correlate to malicious activity within the network. GNNs have been used to identify these anomalies on individual nodes and within paths to detect malicious processes, or on the edge level to detect lateral movement.

Graph attention network -- Part 12

GNNs have become a primary tool for modeling atomic interactions in computational chemistry and materials science, where molecules and crystal structures are naturally represented as graphs with atoms as nodes and chemical bonds or spatial proximity as edges. A key advantage over traditional quantum chemistry methods such as density functional theory (DFT) is the reduction in computational cost from $O(n^3)$ to $O(n)$, enabling simulations at previously impractical scales. Early GNN architectures for this domain include SchNet (2017), which

introduced continuous-filter convolutional layers for learning on 3D molecular structures, and DimeNet (2020), which incorporated angular information between atoms to improve geometric expressivity. Subsequent equivariant architectures such as NequIP and MACE, which respect the rotational and translational symmetries of physical systems, achieved significant accuracy improvements on standard benchmarks including QM9, MD17, and the Materials Project. Large open datasets have driven rapid progress in this area. The Open Catalyst 2020 (OC20) dataset, comprising over 260 million DFT calculations of adsorbate-catalyst surface interactions, established a widely used benchmark for energy and force prediction tasks relevant to catalyst discovery.. The Materials Project and Alexandria databases have similarly provided large-scale crystal structure data for training universal interatomic potentials. More recent datasets including OMat24 and OMol25 have extended coverage to organic molecules and complex materials at greater computational scale. These datasets have enabled the development of universal machine-learning interatomic potentials (MLIPs) — models trained across diverse chemical domains rather than on system-specific data. Examples include CHGNet, M3GNet, MACE-MP, and Meta FAIR's Universal Models for Atoms (UMA), the latter trained on approximately 500 million atomic structures spanning molecules, materials, and catalysts. GNoME, developed by Google DeepMind, applied GNNs to predict the stability of crystal structures, reporting the identification of millions of new stable candidate materials.